Light Collection in a Polished Plastic Scintillator Read Out by Side-Mounted SiPMs: a Geant4 Matrix Simulation Study

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Abstract

We describe a Geant4 simulation of a 250 mm×250 mm×10 mm polished plastic scintillator (PVT, UPS-92S/EJ-200 class) read out by side-coupled silicon photomultipliers (SiPMs), modelled on the muon-trigger detector of Rao *et al.* [J. Appl. Phys. **138**, 114504 (2025)]. The detector is exposed to a vertical 4 GeV muon pencil beam. We present the geometry and the optical-physics implementation in detail—the Geant4 *unified* surface model, the finish/type combinations used to represent four wrapping configurations (bare/black holder, TiO_2 paint, aluminium foil, glossy paper), the photocathode efficiency model that drives photodetection, and the per-event optical accounting that feeds the analysis. A 48-file matrix scan (4 SiPM layouts × 4 reflectors × 3 beam positions, 1000 events each) yields collection efficiencies that reproduce the experimentally observed ordering bare $\ll$ TiO_2 $<$ Al $\approx$ glossy, with the absolute scale set by the SiPM photocathode. We report per-event photon yields, photoelectron spectra, wavelength distributions, and position-resolved efficiency maps. Two implementation caveats are documented: a rare intermittent multi-threaded stall in the optical tracking, and a reversed centre/edge/corner trend in the TiO_2 configuration.

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1 Introduction and motivation

Plastic scintillators read out at their edges by SiPMs are attractive for large-area, low-cost muon triggers and veto systems for cosmic-ray muon tomography [18, 19]. The amount of scintillation light that reaches the photosensor is governed almost entirely by how the non-instrumented faces of the plate are treated: a polished plate in air traps light by total internal reflection (TIR) and can be a nearly lossless light guide, whereas a plate wrapped in an absorbing or diffusive material loses photons at every bounce. Understanding this trade-off quantitatively is the central design question, and it is precisely where simulation is most useful, because the photon transport is a multi-bounce random walk that is difficult to model analytically [7].

This report documents a Geant4 (v11.3.2) [2, 3] simulation built to address that question for the specific geometry and surface treatments of Rao *et al.* [1]. The deliverable is a self-contained "matrix scan": every combination of four SiPM layouts and four reflector treatments, at three beam positions on the plate. The goal is not to fit a single number but to verify that the simulation reproduces the *ordering* of light-collection efficiency that the experiment observes, and to expose the physics behind it.

The remainder of this document is organised as follows. Section 2 describes the detector model: the scintillator, the SiPM read-out, the four read-out layouts (including the wavelength-shifting fibre clamp of config 4), the reflector treatments, and the optical-physics implementation, all verified line-by-line against the Geant4 source. Section 3 presents the matrix simulation—its beam and patch setup, and the resulting collection efficiencies, photon/photoelectron spectra, and position-resolved maps. Section 4 presents the complementary whole-slab irradiation, which samples the full plate uniformly. Section 5 documents known caveats, and Sec. 6 summarises.

2 Detector model

2.1 Scintillator plate

The active volume is a rectangular PVT scintillator $250 \times 250 \times 10$ mm, mirroring the UPS-92S plate of Ref. [1]. It is assigned the Geant4 NIST base material `G4_PLASTIC_SC_VINYLTOLUENE` with a custom optical properties table built from the EJ-200 and UPS-923A data sheets [11, 12, 13]. The optical constants—the refractive index ($n = 1.58$, essentially flat across the visible), light yield, emission peak, bulk attenuation length, scintillation decay and rise times, and the Birks constant [14, 15]—are collected in the consolidated parameter table of Sec. 2.5 and are not repeated here. Two entries are deliberate modelling choices rather than quoted numbers: the bulk attenuation length is peaked at 3.8 m but falls steeply in the UV to model wavelength-shifter self-absorption, and the emission spectrum (Fig. 3) is essentially zero below 380 nm and above 550 nm. Both scintillation and Čerenkov photons are produced and tracked (Sec. 2.7); the optical grid spans 310 nm to 700 nm so that the Čerenkov continuum, which extends well below the scintillation band, is covered.

2.2 SiPM readout and coupling

Each SiPM [8] is modelled mechanically as a $4 \times 4 \times 0.6$ mm box of "SiPM epoxy" ($n = 1.55$), coupled to a side face of the plate through a 0.1 mm silicone grease pad ($n = 1.46$). The cou-

pling chain is therefore *scintillator* → *grease* → *SiPM*, with Fresnel transmission at each dielectric interface and photodetection at the final grease/SiPM boundary.

Four layouts are studied (schematic in Fig. 1):

Config 1 (8 SiPMs) two SiPMs per side face, at ±50% of the half-length.

Config 2 (4 SiPMs) one SiPM per side face, at the *same* ±50% offsets as config 1—i.e. config 1 with one device removed per side, the removed slot rotated from side to side so the layout remains four-fold symmetric.

Config 3 (2 SiPMs) both SiPMs on the *same* face—the top ($+y$) edge—at ±50%, matching T1/T2 of Ref. [1].

Config 4 (1 SiPM + WLS clamp) a single SiPM reading out a wavelength-shifting (WLS) fibre clamp along the $-x$ narrow face (Sec. 2.4).

SiPM footprints are clamped so they cannot overhang the slab face.

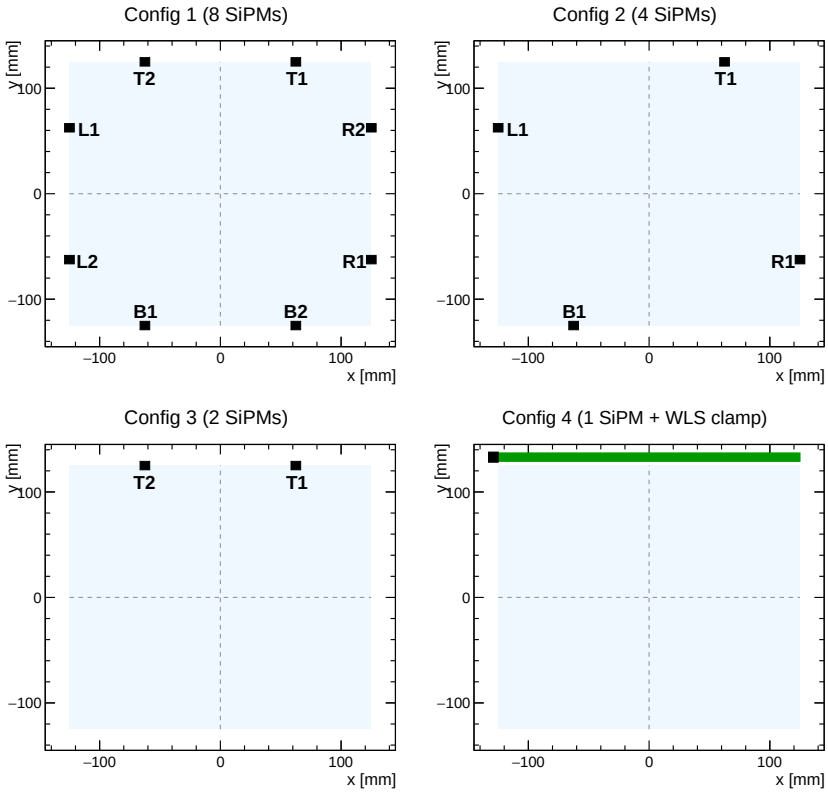


Figure 1: Schematic plan view of the four detector layouts on the 250 mm×250 mm plate footprint. Black squares mark the side-mounted SiPMs (drawn oversized for visibility; the active area is 4 mm×4 mm), and the dashed grey lines are the $x = 0$, $y = 0$ axes. Config 1 places two SiPMs on each of the four edges; config 2 places one SiPM per edge at the same offsets as config 1 (one device removed per side); config 3 places both SiPMs on the top ($+y$) edge; config 4 uses a single SiPM reading out a WLS fibre clamp, drawn along the top ($+y$) edge with the black read-out SiPM at its left end (green bar), following Fig. 3(b) of Ref. [1]. The inwards-offset labels name the eight side-mounted sensors (T/B/R/L = top/bottom/right/left); on each edge "1" marks the sensor nearer the corner reached when travelling clockwise around the perimeter, so "1" sits at the top-right (T1), bottom-right (R1), bottom-left (B1) and top-left (L1) corners. This is the scheme used in the light-sharing discussion below. The same black-square markers are overlaid on the collection-efficiency maps (Figs. 14–17).

2.3 Reflector treatments

Four treatments are applied as a *logical skin* over the non-instrumented surfaces of the plate; the grease/SiPM patches override the skin locally with dedicated *border* surfaces. The mapping from the physical treatment to the Geant4 optical-surface parameters is the heart of the model and is

summarised in Table 1 and justified in Sec. 2.7; the strong sensitivity of light collection to the wrapping and reflector choice is well established experimentally [20].

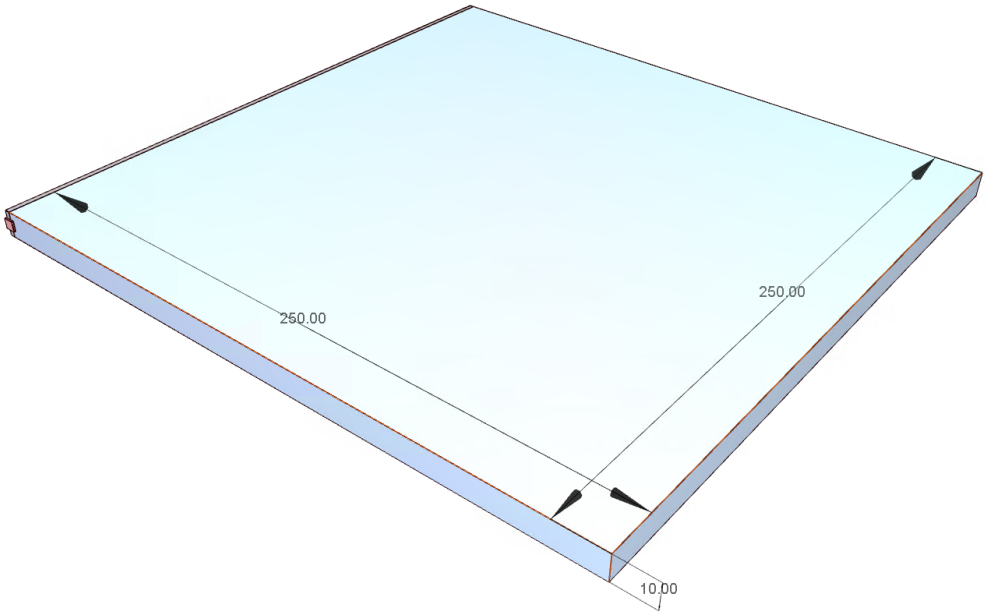
Table 1: Reflector optical-surface model. All surfaces use the *unified* model. $R(\lambda)$ is the tabulated reflectivity over 360 nm to 650 nm.

Config	Geant4 type	Geant4 finish	Key parameters
Bare / black holder	`dielectric_dielectric`	`polished` (absorber)	$R = T = 0$ (perfect sink)
TiO_2 paint	`dielectric_dielectric`	`groundfrontpainted`	Lambertian, $R \approx 0.955$–0.983
Aluminium foil	`dielectric_dielectric`	`polishedbackpainted`	specular spike = 1, $R \approx 0.84$–0.91
Glossy paper	`dielectric_dielectric`	`polishedbackpainted`	specular spike = 1, $R \approx 0.90$–0.955

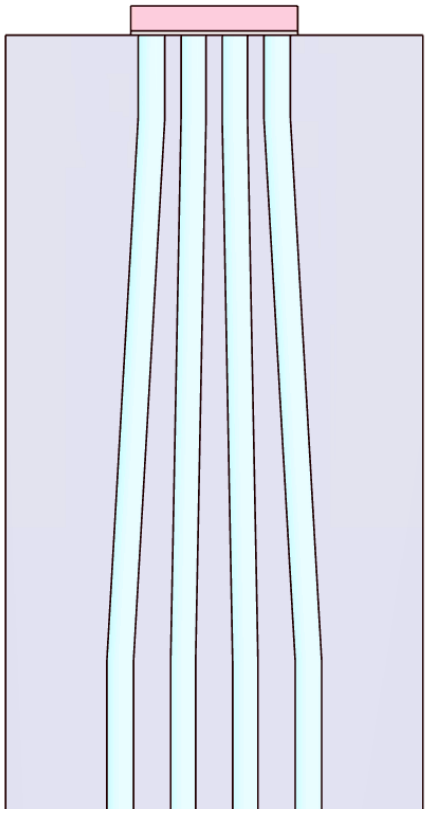
A subtle but essential point concerns the *bare* configuration. A free-standing polished plate is an excellent TIR light guide and would be the *best* collector—contradicting the measurement, in which the bare plate gives the weakest signal. The resolution (Sec. 2.7) is that the bare plate is clamped into a black 3D-printed holder whose refractive index ($n \approx 1.55$) nearly matches the scintillator's. The faces are then in *effective optical contact* rather than in contact with air: TIR is *frustrated*, and light leaks out of the plate into the black walls, where it is absorbed. This is modelled by a perfect-absorber skin on the plate ($R = T = 0$) for the bare case.

2.4 Config 4: wavelength-shifting fibre clamp

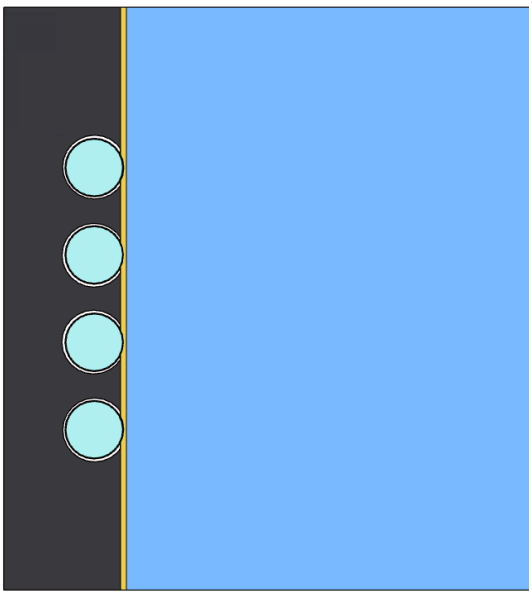
The fourth layout follows the single-SiPM, WLS-readout variant of Ref. [1]. Four 1.0 mm-diameter Kuraray Y-11(200) wavelength-shifting fibres [16, 17] run the full 250 mm length of the $-x$ narrow face in a single row. Over most of the length they are parallel at a 1.5 mm centre-to-centre pitch—so a solid web of clamp material separates the milled 1.1 mm channels—and converge over the last 15 mm to a touching 1.0 mm bundle that presents all four ends to a single 4 mm×4 mm SiPM at the $-y$ end; their far $+y$ ends are closed by individual aluminium mirror caps. A thin 0.10 mm grease film couples the plate face to the clamp, and each fibre protrudes 0.05 mm through its channel mouth into that film, keeping the grease continuous against the scintillator. The deliberate 0.05 mm air annulus between the 1.0 mm fibre and its 1.1 mm channel is essential: the modelled cladding ($n = 1.49$) exceeds the grease ($n = 1.46$) only marginally, so a snug fit against the $n \approx 1.55$ clamp would frustrate core–cladding total internal reflection. The outboard face of the black clamp carries a mirror skin. Because the core–cladding index step is small, guiding relies on the cladding staying optically isolated from the clamp, which the air annulus guarantees. The WLS process is enabled automatically when this geometry is built. In the report's top-down frame the clamp runs along the *top* edge with the single read-out SiPM at the *top-left* corner, as in Fig. 3(b) of Ref. [1].



(a) Three-dimensional view of the complete config-4 assembly, with the principal dimensions annotated.



(b) Top view of the black fibre clamp: the four fibre channels and the read-out SiPM at the end of the converged bundle (report-frame top-left).



(c) Cross-section through the clamp: the scintillator, the thin grease film in the slab–clamp gap, the four fibre discs seated in their 1.1 mm channels with a 0.05 mm air annulus, and the outboard black clamp.

Figure 2: Geometry-only views of the config-4 WLS-fibre clamp, rendered from the Geant4 VRML geometry export (*no* beam-on). Following the event-display colour scheme, fibre cores are pale green, cladding dark green, the grease film amber, the black clamp and holder dark grey, and the read-out SiPM red. The four 1.0 mm fibres sit on the $-x$ narrow face in 1.1 mm channels separated by clamp webs, with only a thin grease film in the slab–clamp gap; they converge to a bundle read out at $-y$ by one SiPM and mirrored at $+y$.

2.5 Summary of detector parameters

The principal parameters of the three active components—the scintillator plate, the WLS fibre, and the SiPM—are collected in Table 2, alongside the corresponding values quoted for the reference experiment of Ref. [1]. Where a quantity is not reported in Ref. [1] or is not modelled here, a dash is shown; the simulation follows the experimental values wherever they are available, and the optical model of Sec. 2.7 is built entirely from the values in the last column.

Table 2: Principal parameters of the scintillator, the WLS fibre, and the SiPM, comparing the reference experiment of Ref. [1] with the values adopted in this simulation.

Parameter	Ref. [1]	This simulation
Scintillator (UPS-92S / PVT)		
Material	UPS-92S (AdvTECH UK)	PVT (`G4_PLASTIC_SC_VINYLTOLUENE`)
Dimensions	$250 \times 250 \times 10$ mm	$250 \times 250 \times 10$ mm
Refractive index	1.58	1.58
Density	1.05 g cm^{-3}	1.032 g cm^{-3}
Light output	50–60% of anthracene (≈ 8 to 10×10^3 ph/MeV)	1.0×10^4 MeV^{-1}
Emission peak	425 nm	425 nm (2.917 eV)
Bulk attenuation length	2000 mm	3.8 m (peak, falling in the UV)
Scintillation decay time	2.4 ns	2.1 ns
Rise time	—	0.9 ns
Birks constant	—	0.126 mm MeV^{-1}
Surface finish	all sides polished	all sides polished
WLS fibre (Kuraray Y-11(200))		
Fibre diameter	1.0 mm	1.0 mm (core+clad)
Cladding	multi-cladding	single PMMA clad ($n = 1.49$)
Core refractive index	—	1.59
Absorption peak	430 nm	430 nm to 460 nm (WLS onset)
Emission peak	476 nm	476 nm
Attenuation length	> 3500 mm	$\gtrsim 5$ m (transport)
WLS decay time	—	7 ns
Read-out arrangement	single row, 250 mm long	4 fibres, 1.5 mm pitch $\rightarrow$ 1.0 mm bundle
Channel diameter	—	1.1 mm (0.05 mm air annulus)
SiPM (CEFLA S.C. ASD-NUV4S-P-40)		
Active area	4×4 mm^2	4×4 mm^2
Number of cells	9340	—
Peak sensitivity	420 nm	420 nm to 425 nm
Photon detection efficiency	43%	0.43 (peak)
Epoxy refractive index	1.5115	1.55
Optical coupling	BC-630 grease	0.1 mm grease film ($n = 1.46$)
Gain / dark-count rate / capacitance	10^6 / 50–100 kHz/mm^2 / 0.84 nF	— (light transport only)

2.6 Optical property tables

Figure 3 plots the three sets of optical constants actually fed to Geant4, over the same 310 nm to 700 nm range used in the transport. The left panel compares the normalised emission spectra: the scintillator peaks at 425 nm, while the Y-11 fibre re-emits in the green, peaked near 480 nm—the shift that makes the config-4 read-out spectrum green. The middle panel gives the absorption (attenuation) lengths on a logarithmic scale: the scintillator is transparent across the visible but its bulk attenuation collapses in the UV (the wavelength-shifter self-absorption band), whereas the fibre is effectively transparent red-ward of 500 nm and absorbs strongly below 480 nm (the WLS onset, marked by the dashed line). The right panel is the SiPM photocathode efficiency = PDE(λ), peaking at 0.43 near 420 nm. These curves are exactly the tabulated points of `src/GeometryMuonScint.cc`; Geant4 interpolates them linearly, so the drawn segments are

what the simulation uses.

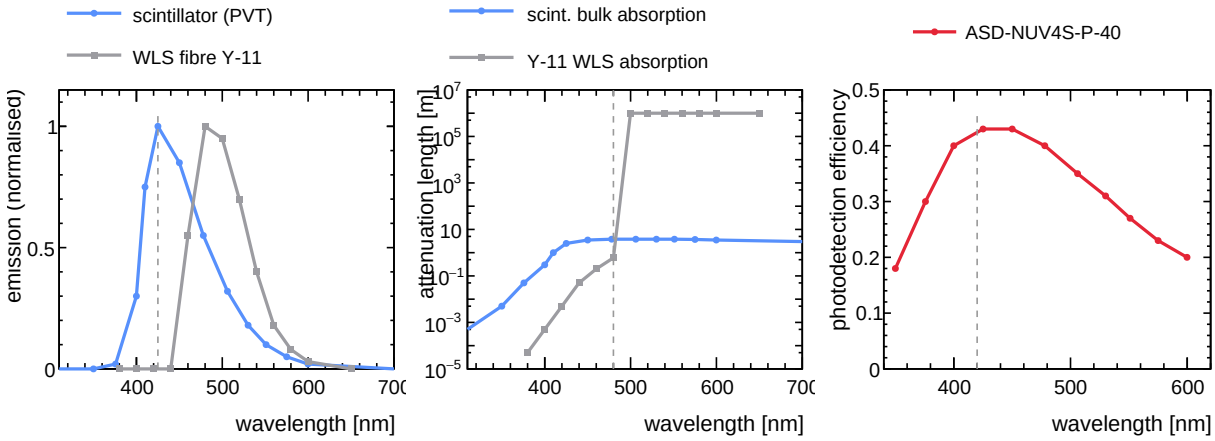


Figure 3: Optical property tables used by the simulation. *Left:* normalised emission spectra of the PVT scintillator (peaking at 425 nm) and of the Y-11 WLS fibre (peaking at 480 nm). *Middle:* absorption (attenuation) lengths on a logarithmic scale—the scintillator bulk absorption and the fibre WLS absorption. *Right:* SiPM photocathode PDE(λ), peaking at 420 nm. The vertical dashed line in each panel marks the characteristic wavelength of that panel's process—scintillator emission peak at 425 nm (left), WLS absorption onset at 480 nm (middle), and SiPM PDE peak at 420 nm (right); the left and right lines therefore nearly coincide, whereas the middle line is red-shifted to the WLS band. Markers are the tabulated points; segments are the linear interpolation Geant4 applies.

2.7 Optical physics implementation

Optical photons are transported by the standard Geant4 optical physics module [4], with scintillation, Čerenkov, bulk absorption, Rayleigh scattering and boundary processes all active. Multiple scattering (Mie) is disabled; wavelength shifting (the `G4OpWLS` process) is enabled only for the config-4 fibre layout (Sec. 2.4) and inert for all others. This subsection describes the boundary physics and the photodetection model, and states explicitly the Geant4 behaviour that each modelling choice relies on, verified against the Geant4 source in `.refs/source/processes/optical`.

2.8 The boundary process and the unified surface model

Photon crossing of a dielectric interface is handled by `G4OpBoundaryProcess`, which selects an `G4OpticalSurface` from the unified surface model of Levin and Moisan [5] (whose key parameters for Geant4/GATE photon transport were later characterised [6]), using (*i*) a `G4LogicalBorderSurface` between two physical volumes or (*ii*) a `G4LogicalSkinSurface` on a logical volume. Reading the source (lines 319–347), the lookup is: a border surface between the pre- and post-step volumes is consulted first; if none exists, the skin surface of the daughter (or, failing that, the mother) logical volume is used. This establishes two facts on which the geometry relies:

1. a `G4LogicalBorderSurface` takes precedence over a `G4LogicalSkinSurface` at the same boundary—hence the grease and SiPM patches punch "holes" in the wrapping skin;
2. if no surface object exists at a boundary, the process falls back to the default lossless TIR behaviour (via `DielectricDielectric()`).

The boundary process requires a valid `RINDEX` on both media; if the property is absent it sets status `NoRINDEX` and kills the photon (lines 289–306, 436–453). Every material in the model therefore carries an explicit `RINDEX` table.

For the *unified* model the probability weights for the reflection sub-models are read from the surface MPT—`SPECULARSPIKECONSTANT` (f_{ss}), `SPECULARLOBECONSTANT` (f_{sl}) and `BACKSCATTERCONSTANT` (f_{bs})—together with `REFLECTIVITY` (line 381) and, where applicable,

`TRANSMITTANCE` (line 394) and `EFFICIENCY` (line 390). The surviving probability $1 - f_{ss} - f_{sl} - f_{bs}$ is shared between diffuse (Lambertian) and bare reflection.

2.9 Reflector finishes

The four treatments use three distinct code paths in `G4OpBoundaryProcess`, which we verified:

- `groundfrontpainted` (TiO_2). The finish is handled by the "painted-front" branch (lines 462–488): a uniform random draw against `REFLECTIVITY` decides absorbed / transmitted / reflected, and reflection is set to a pure Lambertian distribution (`LambertianReflection`, line 481). There is *no* TIR at this finish—photons are either diffusely reflected or absorbed. This is the physically correct description of sub-surface scattering in a thick TiO_2 paint.
- `polishedbackpainted` (Al, glossy paper). This finish is routed to `DielectricDielectric()` (line 456–459). The scintillator/air-gap interface still exhibits Fresnel reflection and TIR; sub-critical photons refract into the gap and are returned by the backing "paint" with reflectivity $R(\lambda)$ and a reflection distribution set by f_{ss}, f_{sl}, f_{bs}. We choose $f_{ss} = 1$ (pure specular spike) for aluminium, since a smooth metal foil is a mirror, and likewise for glossy paper—the experimental finding that Al $\approx$ glossy and both are the brightest means glossy paper is *not* behaving as a diffuser, so it is modelled as a slightly lossier mirror than Al.
- **Bare absorber**. This is deliberately *not* a wrapping skin. Instead a skin with $R = T = 0$ on the plate (and a matching skin on the holder shell) models the frustrated-TIR loss into the black holder.

2.10 Wavelength shifting

In the config-4 layout, photons that reach the fibre cores are absorbed and re-emitted isotropically at longer wavelength by the `G4OpWLS` process, which is configured through the fibre material's `WLSABSLENGTH`, `WLSCOMPONENT` and `WLSTIMECONSTANT` properties (Y-11 absorption peaks near 430 nm, re-emission near 490 nm). The process is switched on by the physics list when the WLS geometry is detected (the Geant4 optical-physics constructor reads the process activation at build time, so the flag is set before the base construction). Because the shifted emission no longer overlaps the fibre absorption band, the re-emitted light is transported losslessly along the fibre by TIR to the read-out SiPM—which is why the detected config-4 spectrum is green-shifted (mean $\sim$485 nm) relative to the direct-coupled configurations.

2.11 Photodetection: the SiPM photocathode

Photodetection is delegated entirely to Geant4, rather than being drawn manually in the sensitive detector. The grease/SiPM boundary is a `dielectric_metal` surface whose MPT carries the SiPM `EFFICIENCY` as a function of photon energy, i.e. the photon detection efficiency (PDE) of an ASD-NUV4S-P-40 device [10], peaking at ≈ 0.43 near 420 nm, with `REFLECTIVITY` $= 0$.

The mechanism is worth stating precisely because it is the reason the sensitive detector needs no random draw of its own. In `DielectricMetal()` (lines 724–817), when the Fresnel–PDE roll places a photon in the "absorbed" channel, `DoAbsorption()` is called. Inside it (header, lines 305–320):

$$f_{\text{status}} = \begin{cases} \text{Absorption}, & \xi \geq \text{EFFICIENCY}(\lambda), \\ \text{Detection}, & \xi < \text{EFFICIENCY}(\lambda), \end{cases} \qquad \xi \sim U(0,1), \tag{1}$$

where EFFICIENCY(λ) is the tabulated PDE. The absorbed photon is assigned a non-zero `LocalEnergyDeposit` *only* when the PDE draw succeeds. Back in `PostStepDoIt`, if the status is `Detection` *and* the invoke-SD flag is set, the sensitive detector is invoked (lines 559–560):

$$f_{\text{status}} = \text{Detection} \ \wedge \ f_{\text{InvokeSD}} \implies \text{InvokeSD(step)}. \tag{2}$$

The flag is enabled in the physics list via `G4OpticalParameters::SetBoundaryInvokeSD(true)`. Consequently the sensitive-detector `ProcessHits` fires *only* for genuine photoelectrons (it tests that the step energy deposit is positive), counts one N_{pe} per call, and kills the photon (the SiPM is opaque). This is a clean separation of concerns: the physics of the photocathode lives in the optical surface, and the sensitive detector only books hits.

2.12 Per-event optical accounting

A thread-local diagnostic record (`OpticalDiag`) accumulates, per event: scintillation photons, Čerenkov photons, terminated photons, detected photoelectrons, and photons that escaped to the world. The primary muon's first-entry position in the slab is captured on the first step inside the scintillator logical volume and stored in slab-local coordinates. At end of event the sensitive detector writes one row per (event, SiPM), *including zero-detection rows*, so the sample is unbiased in the muon interaction position. Each row carries the entry position, the event's produced-photon count (the efficiency denominator), the detected count in that SiPM, the first arrival time, the mean and RMS detected wavelength, and the track weight. This one tree supports per-SiPM, combined, and spatially-binned collection-efficiency analyses.

Figure 4 shows a head-less event display of a single vertical muon in the glossy-wrapped plate (config 1). For legibility the display uses a 30 MeV muon—just below the Čerenkov threshold of the scintillator—with the scintillation yield rescaled, so that the individual optical tracks can be followed: photons are generated along the muon path, propagate by total internal reflection and specular reflection off the wrapper, and terminate on the SiPM patches at the four edges.

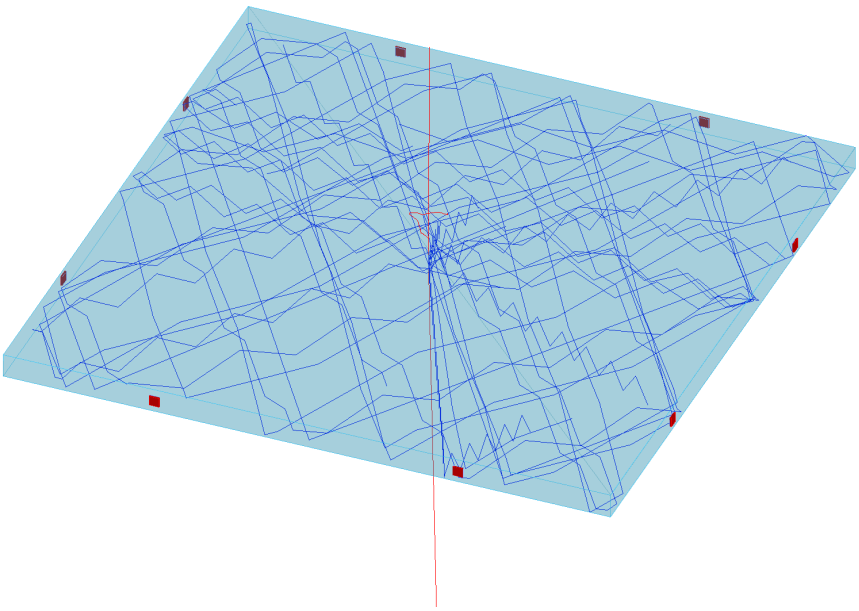


Figure 4: Head-less (`TSG_OFFSCREEN`) event display of one vertical muon in the glossy-wrapped plate with the config-1 layout, on a white background. The red line is the muon and the blue lines are optical photons (here purely scintillation, since the muon is below the Čerenkov threshold); the plate and its reflective wrapper are drawn as translucent surfaces with their edges outlined, and the SiPMs are opaque red. Both the muon energy (30 MeV, below the Čerenkov threshold) and the photon yield are chosen for visualisation only, so that individual tracks are resolved rather than merging into a saturated blob. The display is generated reproducibly by `macros/muon_vis.mac`.

3 Matrix simulation

3.1 Beam, irradiation patches, and efficiency definition

The primary source is a vertical muon pencil beam: a 50 mm×50 mm square beam of 4 GeV μ^- normally incident on the plate, with the beam centre placed at three positions, each irradiating a 50 mm×50 mm patch lying *entirely on* the polished plate. With the plate spanning $[-125, 125]$ mm in both x and y, the patch centres are

$$\text{centre} = (0, 0), \qquad \text{edge} = (0, -100), \qquad \text{corner} = (100, -100)\ \text{mm},$$

so the illuminated regions are $[-25, 25]^2$, $[-25, 25] \times [-125, -75]$ and $[75, 125] \times [-125, -75]$ respectively (report frame: $+y$ is the top, $-x$ the left). This places the edge and corner regions on the bottom ($-y$) edge, opposite the single instrumented face of config 3 (top, $+y$), as in Fig. 3 of Ref. [1]. It also matches the 50 mm×50 mm active area of the muon coincidence detector (MCD) that defines the regions of interest in Ref. [1], and keeps the irradiated area—and hence the denominator of Eq. 3—equal at all three positions. Each (config, reflector, position) runs 1000 events after a single `/run/initialize`. The matrix therefore comprises

$$4\ \text{configs} \times 4\ \text{reflectors} \times 3\ \text{positions} = 48\ \text{files}, \qquad 48 \times 1000 = 48\,000\ \text{events}.$$

Runs used 8 threads on an 8-thread (4-physical-core) workstation. The optional per-slab optical track-length flux map is disabled for the matrix (it is retained as a separate diagnostic); the collection efficiency is reconstructed entirely offline from the `sipmHits` tree at any desired binning, so the map is not needed.

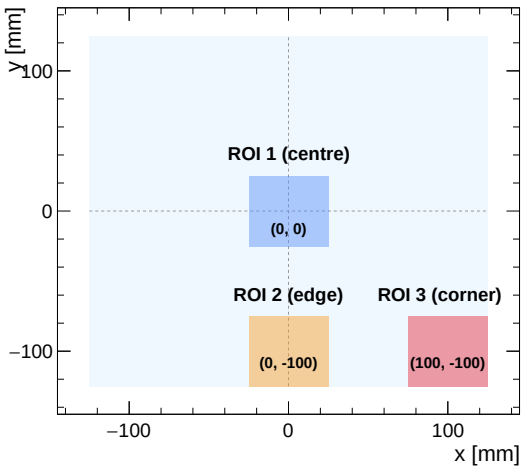


Figure 5: Plan view of the three irradiation sites (regions of interest) on the 250 mm×250 mm plate, in the report frame ($+y$ top, $-x$ left). Each coloured square is the 50 mm×50 mm footprint of the vertical muon beam: ROI 1 (centre, blue) at $(0, 0)$, ROI 2 (edge, orange) at $(0, -100)$, and ROI 3 (corner, red) at $(100, -100)$. The edge and corner sites lie on the bottom ($-y$) edge, opposite the single instrumented face of config 3.

The collection efficiency is defined, for a set $\mathcal{S}$ of SiPMs and an event e, as

$$\varepsilon = \frac{\sum_e \sum_{s \in \mathcal{S}} N_{\text{pe}}^{(e,s)}}{\sum_e N_\gamma^{(e)}}, \tag{3}$$

where $N_\gamma^{(e)}$ is the total number of optical photons generated in the slab during event e (scintillation plus Čerenkov) and $N_{\mathrm{pe}}^{(e,s)}$ is the number of photoelectrons in SiPM s. The all-SiPM sum over the three beam positions is quoted as the figure of merit.

For presentation, all position-resolved results (the maps of Fig. 5 and the whole-slab maps of Sec. 4) are shown in the report frame with the top edge at $+y$, the bottom at $-y$, the left at $-x$ and the right at $+x$. The internal simulation frame is this view rotated by 90° clockwise; the primary muon entry points and the SiPM markers are mapped into the report frame by $(x, y) \to (y, -x)$ before plotting. This rotation is a rigid change of view and leaves every scalar result (yields, spectra, efficiencies) unchanged; it only fixes the orientation of config 3 (top edge) and of the edge/corner irradiation sites (bottom edge).

3.2 Overall collection efficiency

Figure 6 shows the overall collection efficiency (Eq. 3, all SiPMs, all three beam positions pooled) for every (config, reflector) pair, and Table 3 lists the numbers. The ordering

$$\text{bare}\ (7{\times}10^{-6}\text{–}2{\times}10^{-4}) \ll \mathrm{TiO_2}\ (2{\times}10^{-4}\text{–}10^{-2}) < \mathrm{Al}\ (1.3\text{–}2.6{\times}10^{-2}) \approx \text{glossy}\ (1.5\text{–}3.0{\times}10^{-2})$$

holds for all four layouts, reproducing the experimental result of Ref. [1]. The bare configuration is suppressed by two to three orders of magnitude, as required by the frustrated-TIR absorber model: without a reflective wrap, the trapped light is lost to the black holder. Al and glossy paper are statistically indistinguishable within the 1000-event-per-position sample and are the best collectors. Four SiPMs on four faces (config 2) outperform two SiPMs on one face (config 3), and the eight-SiPM layout (config 1) gives the highest efficiency, as expected from the larger photosensitive coverage. The single-SiPM WLS-fibre layout (config 4) collects substantially less than the direct-coupled layouts: only the fraction of light that reaches the fibre face is wavelength-shifted and guided to the sensor, so its efficiency sits roughly a factor of three to six below config 3.

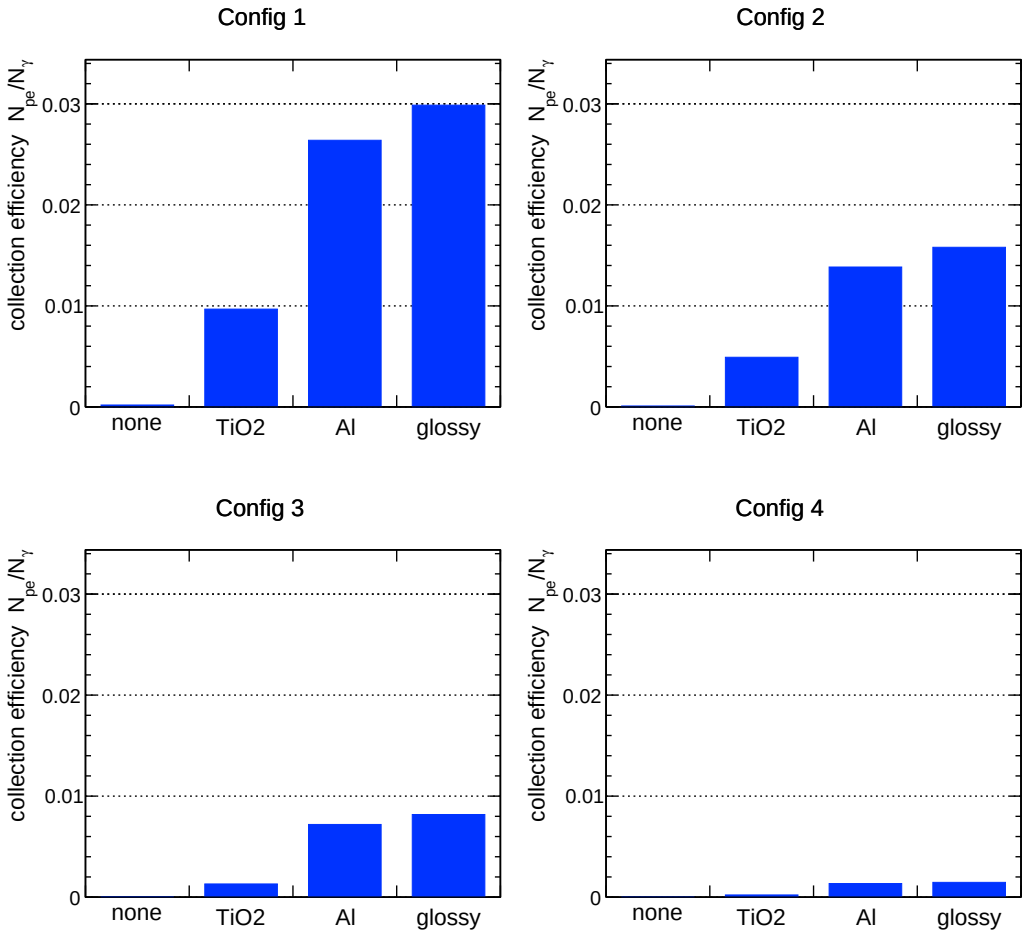


Figure 6: Overall collection efficiency $\varepsilon = \sum N_{\mathrm{pe}} / \sum N_\gamma$ for the four reflector treatments and four detector layouts. The ordering bare $\ll$ TiO_2 $<$ Al $\approx$ glossy is reproduced in every layout. Error bars assume Poisson counting of photoelectrons.

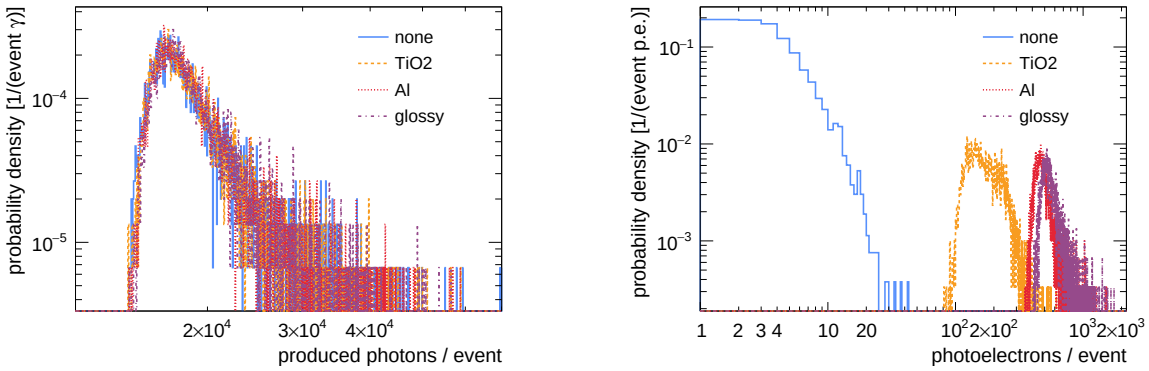
Table 3: Overall collection efficiency (all SiPMs, all beam positions pooled). All beam patches lie fully on the plate.

Layout	Bare	TiO_2	Al	Glossy
Config 1 (8 SiPMs)	1.9×10^{-4}	9.7×10^{-3}	2.64×10^{-2}	2.99×10^{-2}
Config 2 (4 SiPMs)	9.6×10^{-5}	4.9×10^{-3}	1.38×10^{-2}	1.58×10^{-2}
Config 3 (2 SiPMs)	2.4×10^{-5}	1.30×10^{-3}	7.20×10^{-3}	8.18×10^{-3}
Config 4 (1 SiPM, WLS)	6.9×10^{-6}	2.2×10^{-4}	1.35×10^{-3}	1.47×10^{-3}

3.3 Photon yield and photoelectron spectra

Figure 7a shows the per-event optical-photon yield (the denominator of Eq. 3) for the four reflectors. The mean yield is essentially independent of the reflector—about 19 000 photons per event—which is the correct physical behaviour: the reflector changes how photons are transported, not how many are produced. The small spread reflects event-to-event variation in the muon path length.

Figure 7b shows the corresponding photoelectron spectra. Here the reflector has a dramatic effect: the bare configuration peaks near zero with a mean of only 3.7 N_{pe}/event, whereas glossy paper has a mean of 586 N_{pe}/event. The shift from a near-zero-peaked to a broad distribution is the direct signature of effective light collection.



(a) Yield (produced photons/event).

(b) Photoelectrons/event.

Figure 7: Per-event distributions for config 1, plotted as a probability density (unit area, 1 event per unit of the abscissa) with logarithmic x- and y-axes. The production yield (left) is reflector-independent, as it must be; the detected photoelectron count (right) depends strongly on the wrapper. The log y-axis is needed because the four curves have the same event count but very different widths: the concentrated bare peak and the broad reflective distributions cannot both be resolved on a linear scale.

The efficiency scale is set by the SiPM PDE and the coupling: of order 19 000 photons are produced and the best wrapper delivers ≈ 586 photoelectrons, i.e. $\approx 3\%$. Given a PDE of ~ 0.4 and the geometric acceptance of a side-mounted sensor, this is consistent.

3.4 Wavelength of detected photons

Figure 8 shows the wavelength distribution of detected photoelectrons, weighted by their count, for the four reflectors. The distribution peaks near 460 nm for the reflective wrappers (measured mean 459.7 ± 4.8 nm for Al, config 1), i.e. slightly red-shifted with respect to the 425 nm emission peak. This is the expected consequence of self-absorption: the shorter-wavelength part of the emission overlaps the absorption band and is attenuated over the multi-bounce path, so the surviving and detected spectrum is red-shifted. The bare configuration has too few detected photons for a meaningful spectrum.

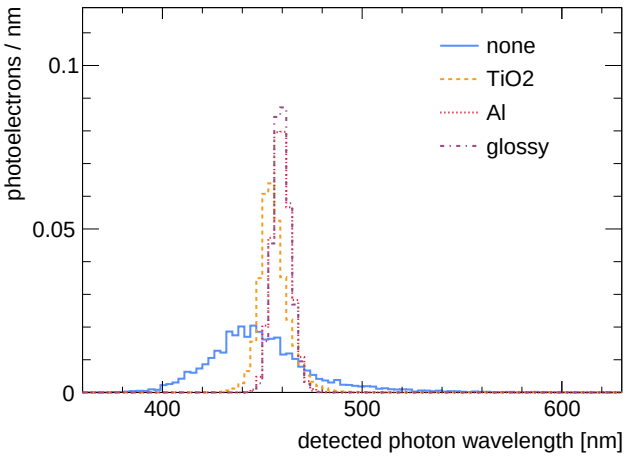


Figure 8: Wavelength distribution of detected photoelectrons (config 1), weighted by detected count. The reflective wrappers give a distribution peaking near 460 nm, red-shifted from the 425 nm emission peak by self-absorption over the multi-bounce path.

Figure 9 compares the detected-photon spectrum of the four configurations for each wrapper. Configurations 1–3, in which the SiPMs view the tile directly, all detect the same blue scintillation spectrum: the *mean* wavelength is essentially independent of the number of sensors, moving only from 459.8 nm (glossy, config 1) to 460.1 nm (config 3), the small increase reflecting the slightly longer collection path of the single-sided layouts. Config 4 is qualitatively different: the WLS fibres absorb the blue light and re-emit it at their own emission band, so the detected spectrum is shifted to 485.2 nm (glossy) — a green-shift of some 25 nm. The same shift appears for every wrapper, including the bare tile (which, with its very small photon count, is statistically noisy), confirming that it originates in the fibre wavelength shifter rather than in the wrapper.

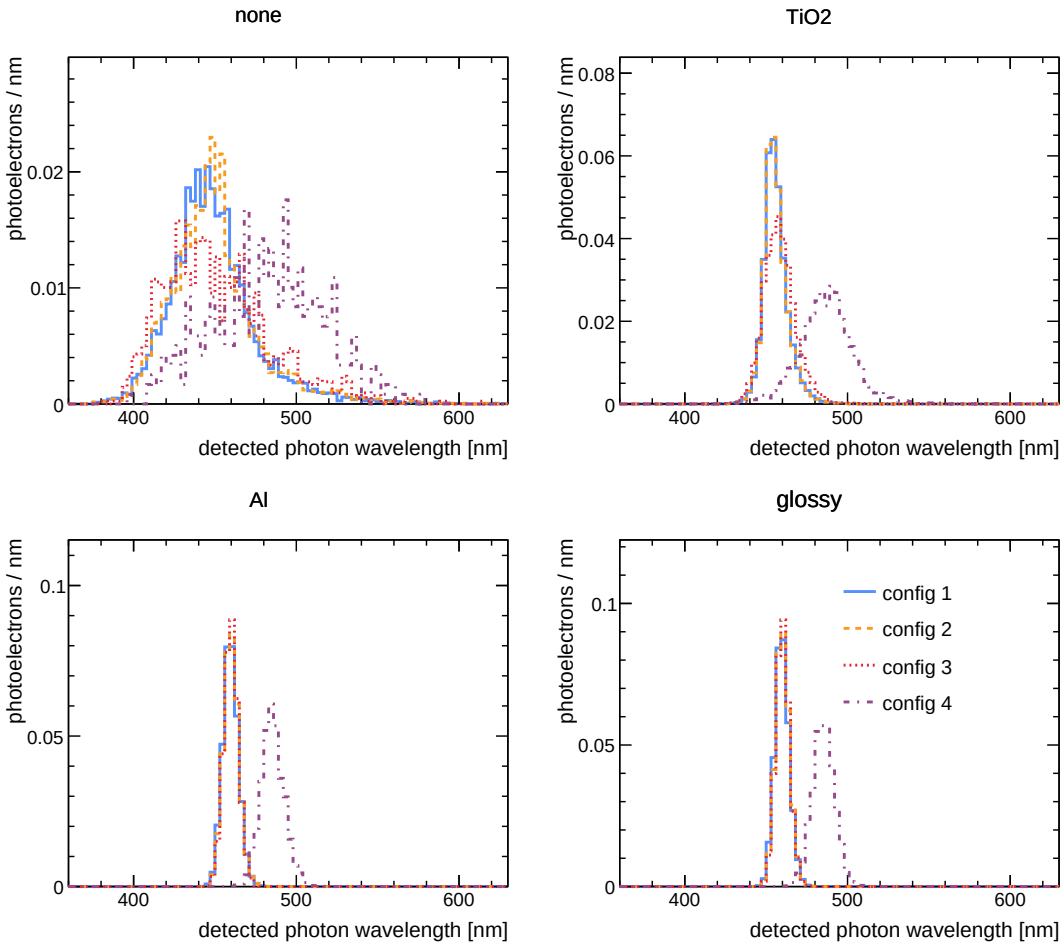


Figure 9: Detected-photon wavelength distribution, weighted by photoelectron count, for each wrapper (panel) and each configuration (curves), using the pooled data of the three irradiation positions. Configurations 1–3 follow the 460 nm blue scintillation spectrum regardless of sensor count, whereas the WLS-fibre readout of config 4 is green-shifted to ≈ 485 nm by the Y-11(200) emission band.

3.5 Spatial and directional dependence

The matrix scan separates the three 50 mm×50 mm patches (centre, edge, corner) precisely so that the response can be resolved by position. The tempting single-coordinate summary is the distance r of the patch from the plate centre, but r is blind to azimuth, and the shortcoming is decisive here: configs 3 and 4 instrument a *single* face, so the collection depends on the *direction* from which the plate is irradiated, not merely on the distance. All three patches also lie on the half of the plate opposite the instrumented face of the single-sided layouts, so their distances are not even monotonic in the proximity to a sensor. We therefore keep the three patches separate and compare them directly.

Figure 10 groups the three patches by layout (one panel per config, four reflector curves) and Fig. 11 groups them by reflector (one panel per wrapper, four layout curves). Because every patch lies fully on the plate and all three have equal area, the produced-photon denominators are equal and the three points within each curve are directly comparable. The curves are plotted on a logarithmic y-axis, so the weakly collecting bare plate is resolved alongside the reflective wrappers.

The conspicuous feature is that the reflective wrappers *homogenise* the response: for Al and glossy paper the efficiency varies by at most 4 % across the three patches in every layout, i.e. an edge- or corner-irradiated plate collects almost exactly as much light as a centre-irradiated one, whereas the diffuse TiO_2 paint varies by a factor $\approx$1.7–2.1 and the bare plate by up to 3.7 (the largest value, config 4, whose single readout amplifies any asymmetry). This is the expected behaviour of a closed reflecting cavity: after the first few bounces the light distribution no longer remembers where the muon crossed the plate, so the sensor coverage, and not the irradiation point, sets the collection.

These three patches lie on the half of the plate *nearest* the sensors, so the comparison above captures only the position dependence within the well-lit region. The response over the *whole* plate is resolved in Sec. 4.2; there the full-plate position contrast is found to be far larger, reaching two orders of magnitude for the bare plate.

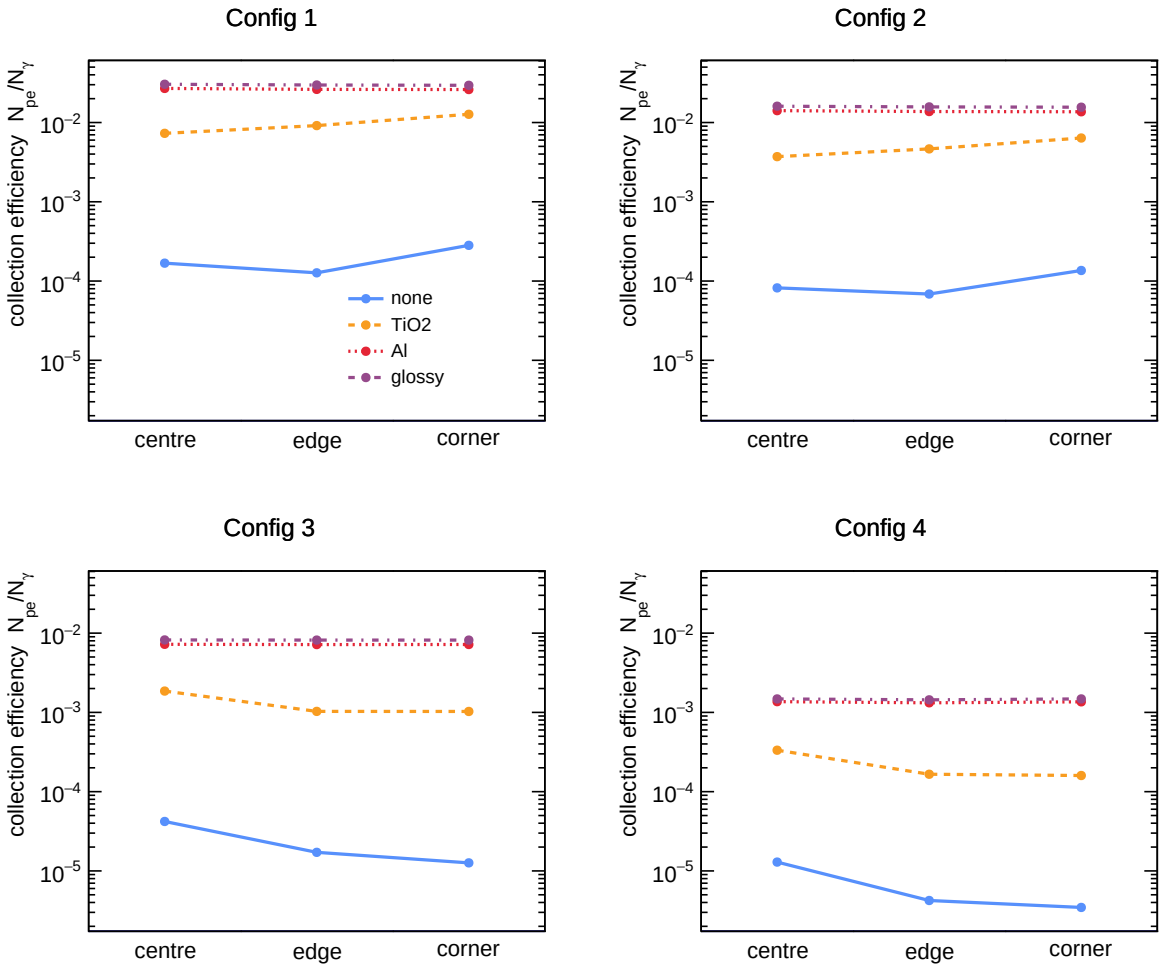


Figure 10: Collection efficiency at the three irradiation patches (centre, edge, corner), one panel per detector layout, with the four reflector curves (logarithmic y-axis). All three patches have equal area and lie fully on the plate, so the points are directly comparable. The reflective wrappers are nearly flat across the three patches in every layout; the bare plate is both weakest and most position-dependent.

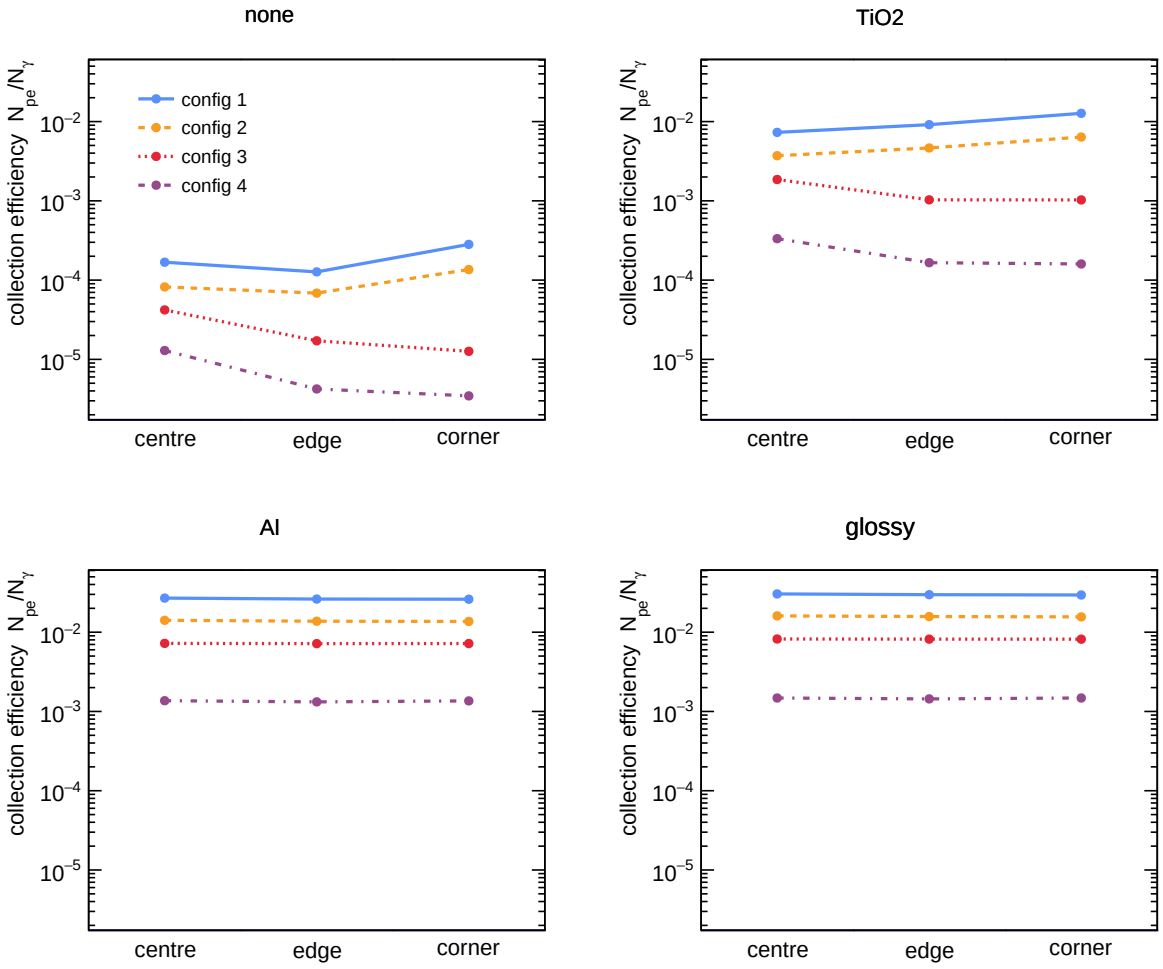


Figure 11: The same data grouped by reflector (one panel per wrapper), with the four layouts as curves. Configs 1 and 2 (sensors on every face) are the most uniform; the single-sided layouts (3 and 4) drop toward the edge and corner patches, which sit on the far side of the plate from their sensors.

Figure 12 resolves the directional response most directly. For config 1 (the layout with the most sensors) it shows the *share* of the collected light seen by each of the eight sensors, in one panel per reflector, with the three irradiation patches as grouped bars (dashed line: the uniform expectation of 1/8). The difference between the wrappers is stark. For the *bare* plate the light barely mixes: at the corner patch the two sensors nearest the beam (B2, R1) together collect 91 % of all photoelectrons, i.e. some 3.6 times the uniform share each, while the remaining six share only 9 % and two of them (B1, R2) record nothing at all. A diffuse or mirror wrapper erases this structure: for glossy paper the same two nearest sensors reach only about 1.1 times the uniform share. The total efficiency is nearly the same for all three patches (Table 4), so the cavity keeps the *amount* of collected light constant while completely rearranging its spatial division. A radius-only summary averages these distinct per-sensor patterns into a single number and hides precisely this information.

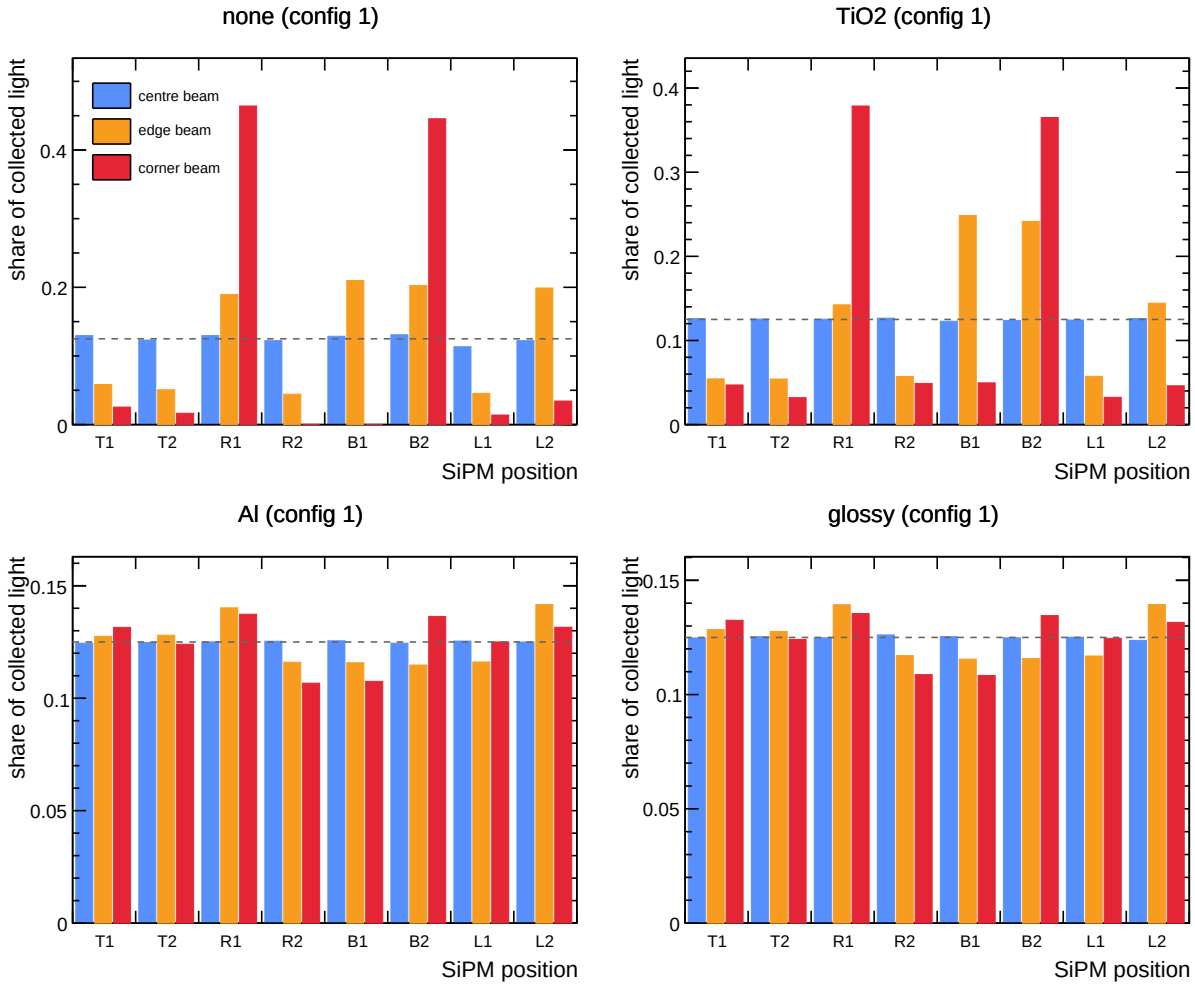


Figure 12: Direction-aware light sharing, config 1, one panel per reflector. Each group of bars is one sensor (x axis) and the three colours are the irradiation patches (blue: centre, orange: edge, red: corner); the height is that sensor's share of the light collected at that patch, and the dashed line marks the uniform expectation 1/8. For the bare plate the light stays concentrated on the two sensors nearest the beam, whereas the reflective wrappers spread it almost uniformly over all eight.

Table 4 lists the absolute centre/edge/corner efficiencies for config 1; the fully position- and direction-resolved maps over the *entire* plate are given in Sec. 4.

Table 4: Position dependence of the collection efficiency, config 1 (all SiPMs). Each beam patch is 50 mm×50 mm and lies fully on the plate, so the produced-photon denominators are equal.

Reflector	Centre	Edge	Corner
Bare	1.7×10^{-4}	1.3×10^{-4}	2.8×10^{-4}
TiO_2	7.3×10^{-3}	9.1×10^{-3}	1.27×10^{-2}
Al	2.69×10^{-2}	2.62×10^{-2}	2.61×10^{-2}
Glossy	3.04×10^{-2}	2.97×10^{-2}	2.96×10^{-2}

3.6 Photoelectron spectra across configurations and positions

Figure 13 collects the per-event photoelectron spectra over the whole matrix scan in the layout of Fig. 10 of Rao et al. [1]: the three irradiation regions (centre, edge, corner) are the rows and the detector configurations are the columns, with the four reflector treatments overlaid in every cell. Two adaptations are made. First, the voltage pulse-height spectra of the measurement are replaced by the simulation's number of photoelectrons per event, the quantity that is directly

comparable between our model and the experiment. Second, the fourth configuration (single SiPM + WLS fibre clamp) is added as a fourth column, so that all four layouts are shown on a common footing rather than in a separate, differently-scaled figure.

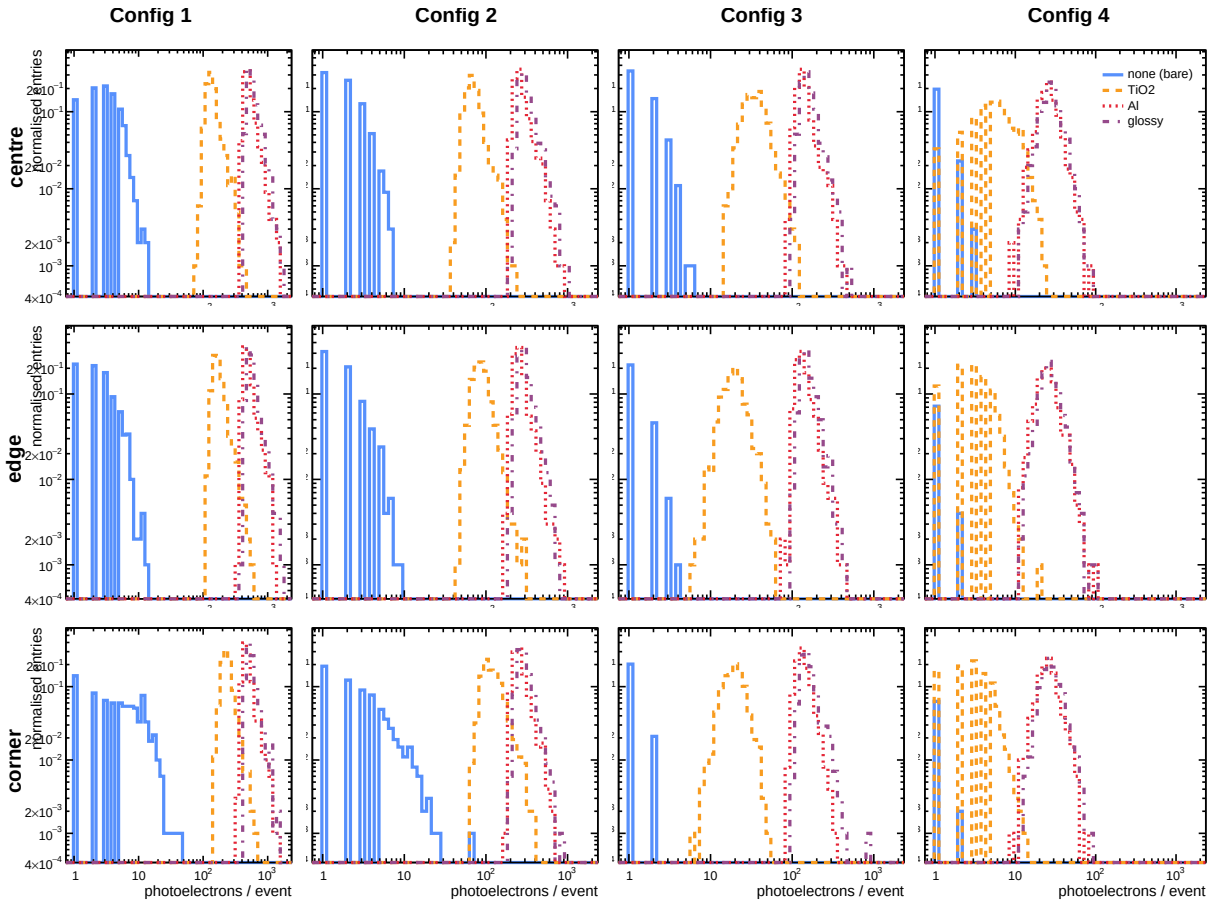


Figure 13: Photoelectron spectra in the layout of Fig. 10 of Rao et al. [1]: rows are the irradiation regions (centre, edge, corner) and columns the detector configurations (1–4, with the WLS-clamp layout as the fourth column). Each cell overlays the four reflector treatments. The histograms are probability densities (unit area on the populated bins) drawn on logarithmic x- and y-axes; zero-photoelectron events form a pulse below threshold and lie off the logarithmic abscissa. The reflective wrappers shift the spectrum by two to three orders of magnitude and broaden it into a recognisable pulse-height distribution, while the bare plate remains a near-threshold peak at all positions and in all layouts.

The figure condenses the trends established above. The bare plate is near-threshold everywhere—its whole spectrum sits below a few photoelectrons—whereas TiO_2, Al, and glossy paper all produce broad, well-separated spectra; at a fixed position the reflector ordering $TiO_2 < Al \lesssim$ glossy is preserved in every cell, mirroring the experimental conclusion that the two bright wrappers are of comparable quality. The position dependence is configuration-specific and not monotone: for the bare and TiO_2 layouts the corner spectrum is shifted highest (a factor ≈ 1.7 in the mean for TiO_2, config 1), for Al and glossy paper the three positions are essentially degenerate, and for the single-sided configurations 3 and 4 the centre is highest and the corner lowest, as expected from their placement opposite the beam patches. This is the same reflector-versus-position interplay summarised in Table 4 and the efficiency maps, presented here as the pulse-height distributions that a threshold-based read-out would actually see.

4 Whole-slab irradiation

4.1 Uniform whole-plate irradiation and full-plate maps

The matrix scan illuminates only three 50 mm×50 mm patches, so it resolves the light collection at the centre, edge and corner but not the full plate. To obtain a complete position-resolved map we ran a second, *whole-slab* configuration: the GPS plane is expanded to 250 × 250 mm (`halfx` = `halfy` = 125 mm), so the primary muon's first-entry position is sampled *uniformly* over the entire plate. Twenty-five thousand events are generated per (config, reflector); the efficiency is reconstructed from the same `sipmHits` tree, in 5 × 5 mm cells, as

$$\varepsilon(x, y) = \frac{\sum_{e \in \text{cell}} N_{\text{pe}}^{(e)}}{\sum_{e \in \text{cell}} N_{\gamma}^{(e)}},$$

i.e. identically to Eq. 3 but binned in the entry position. Because the beam now covers the whole plate, the whole-slab maps fill in the gaps left by the three-patch scan.

Figure 14 shows the complete set of full-plate maps for config 1, one panel per reflector. For the bare plate, averaged over the whole surface the efficiency is $\varepsilon = 4.5 \times 10^{-4}$, consistent with the three-patch bare values of Table 4. The map reveals a clear radial structure that the three discrete patches could only hint at: the efficiency is lowest at the centre ($\varepsilon \approx 1.8 \times 10^{-4}$ within $r < 60$ mm) and rises monotonically toward the edges ($\varepsilon \approx 5.1 \times 10^{-4}$ in the outer region), i.e. by roughly a factor of three. This is exactly the signature of edge readout combined with frustrated-TIR absorption: photons created near a sensor-bearing face have a short path to detection, whereas photons born at the centre must random-walk to an edge through the lossy holder contact, and most are absorbed before arriving.

The reflective panels show the same ordering as the matrix scan, with the position dependence progressively erased. TiO_2 is about twenty times brighter than the bare plate ($\varepsilon = 1.08 \times 10^{-2}$) and its centre-to-edge gradient is much weaker (7.7×10^{-3} for $r < 60$ mm versus 1.15×10^{-2} beyond, a factor of $\sim$1.5). Al is brighter again (2.69×10^{-2}) and almost perfectly *flat*: the centre (2.68×10^{-2}) equals the outskirts (2.69×10^{-2}). Glossy is the brightest (3.04×10^{-2}) and likewise flat (3.03 vs. 3.05×10^{-2}). Diffuse recycling (TiO_2) and specular mixing (Al, glossy) both reduce the penalty of being far from a sensor, the specular mirrors so thoroughly that the entry position is effectively erased, whereas the bare plate—whose frustrated TIR leaks into the holder—retains a steep edge gradient.

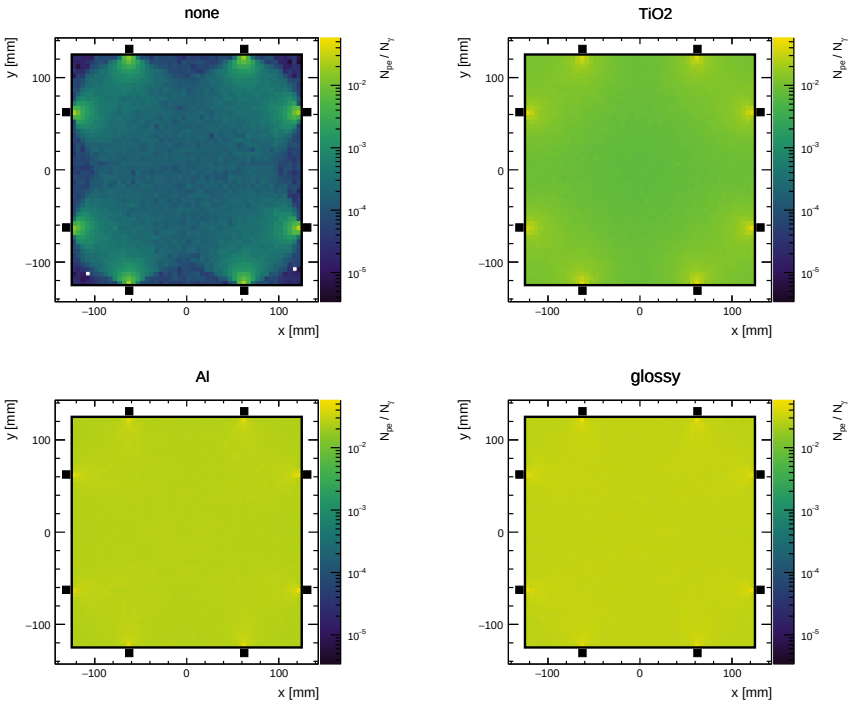


Figure 14: Full-plate collection-efficiency maps for config 1, from 25 000 whole-slab events per panel (5×5 mm cells, on a logarithmic colour scale shared by the panels). Bare plate (top left) rises from the centre toward the instrumented edges; TiO_2 (top right) is roughly twenty times brighter with only a weak gradient; Al (bottom left) and glossy (bottom right) are brighter still and almost flat. The bold rectangle is the 250 mm×250 mm slab outline and the black squares mark the eight SiPM positions.

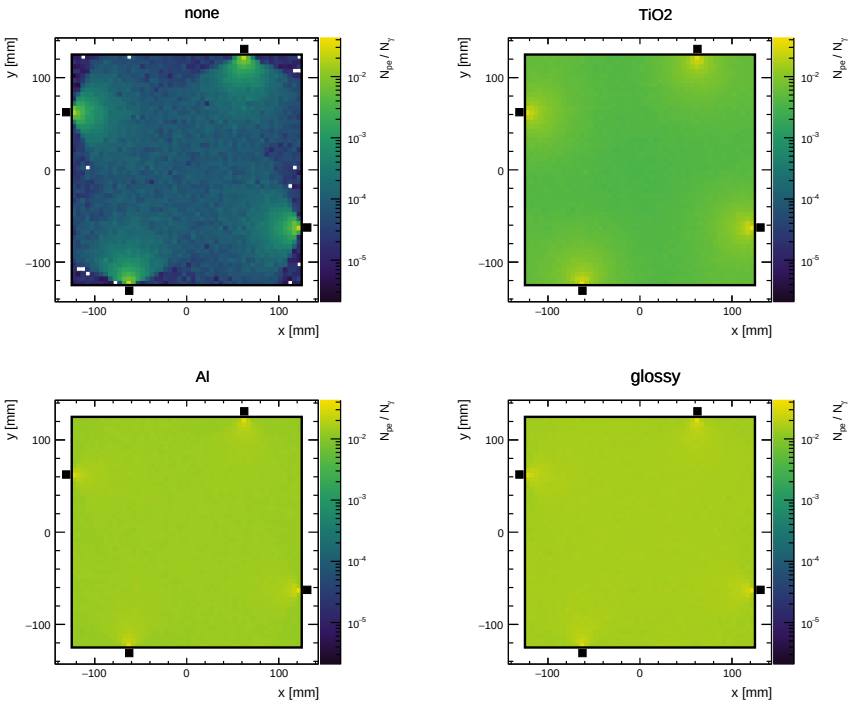


Figure 15: Full-plate collection-efficiency maps for config 2 (four SiPMs, one per lateral face at the config-1 offsets), otherwise identical to Fig. 14. The reflector ordering and the centre-to-edge flattening are the same, with each efficiency about half the config-1 value.

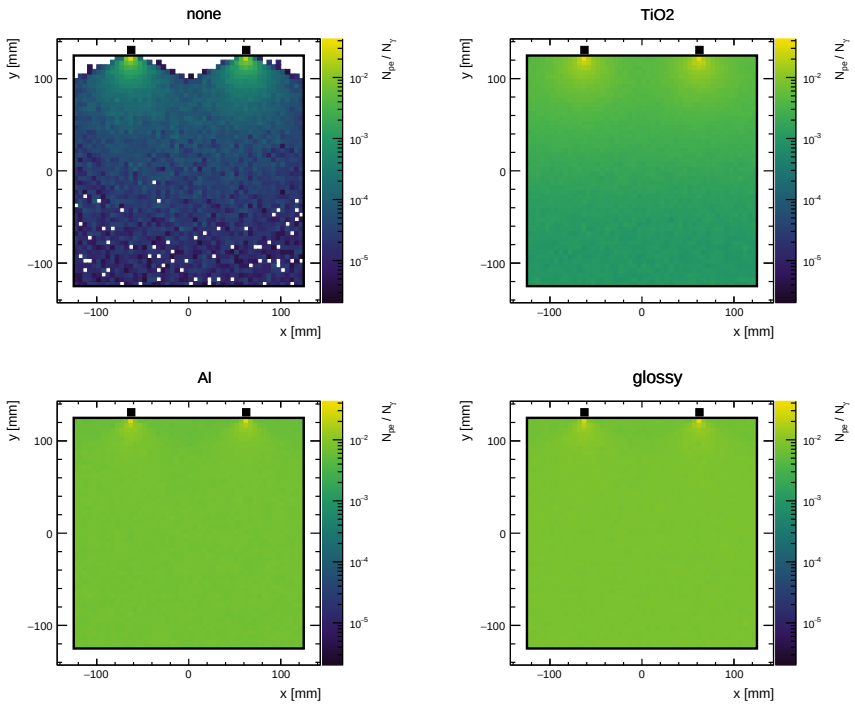


Figure 16: Full-plate collection-efficiency maps for config 3 (two SiPMs, both on the top $+y$ face). The lowest efficiencies of the direct-coupled layouts and the clearest top–bottom asymmetry appear here; the specular wrappers (Al, glossy) again flatten the response.

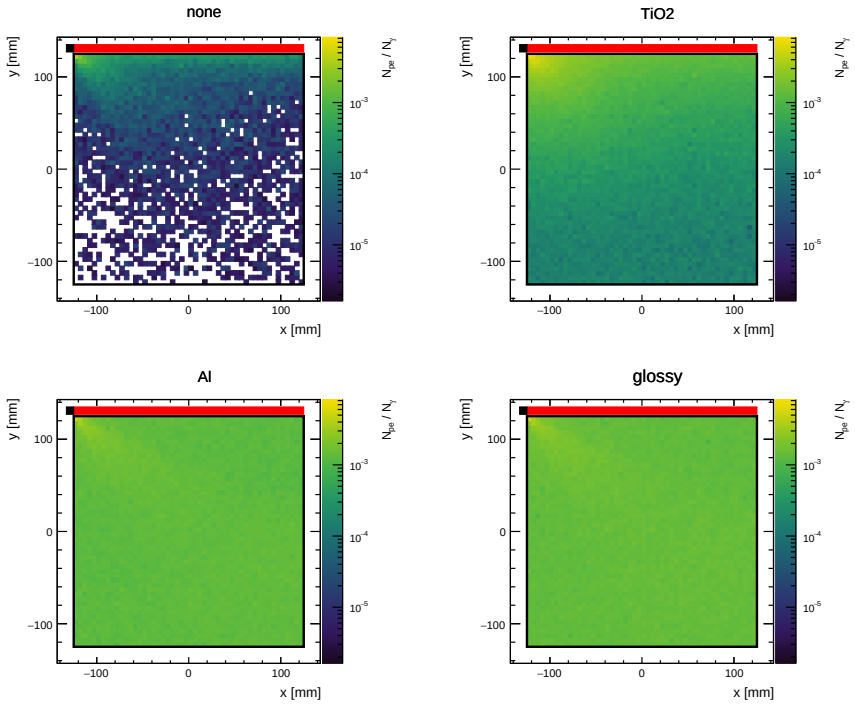


Figure 17: Full-plate collection-efficiency maps for config 4 (single SiPM + WLS fibre clamp). The lowest efficiencies of all four layouts appear here; the bare and TiO_2 panels retain a centre-to-edge gradient, while the specular wrappers (Al, glossy) are almost flat. The red bar along the top ($+y$) edge marks the WLS fibre clamp (the fibres run the full plate length along it) and the black square at its left end is the single read-out SiPM, so the fibres can be located relative to the read-out. Same report frame and binning as Figs. 14–16, each panel on its own logarithmic colour scale.

Figure 15 repeats the whole-slab scan for config 2 (four sensors, one on each lateral face at the config-1 offsets). Every configuration is weaker than its config-1 counterpart—roughly halved, as expected from the reduced sensor area—while the reflector ordering and the flattening behaviour are unchanged. The bare plate gives $\varepsilon = 2.2\times10^{-4}$ with the same strong edge gradient (9.0×10^{-5} at the centre versus 2.5×10^{-4} outside $r = 60\,\mathrm{mm}$); TiO_2 reaches 5.5×10^{-3} with a milder gradient, and Al and glossy reach 1.42×10^{-2} and 1.61×10^{-2}, both nearly flat. The monotone bare $\ll$ TiO_2 < Al $\approx$ glossy sequence and the progressive erasure of the position dependence are therefore robust against the sensor count: fewer sensors lower the overall light collection but do not change the physics of the wrapper.

Finally, Fig. 16 shows config 3, in which both sensors sit on the same (top, $+y$) face. The integrated efficiencies are the lowest of the direct-coupled configurations—bare 1.1×10^{-4}, TiO_2 2.7×10^{-3}, Al 7.2×10^{-3}, glossy 8.2×10^{-3}—because concentrating both sensors on one face halves the solid angle available to photons born on the opposite side. The maps make the resulting asymmetry visible: the bare and TiO_2 panels are brightest near the instrumented top ($+y$) edge and dimmest on the far ($-y$) side (4.5×10^{-5} at the centre versus 1.2×10^{-4} outside $r = 60\,\mathrm{mm}$ for the bare plate). The specular wrappers again homogenise the response, with Al and glossy flat to better than 1% between the inner and outer regions.

Figure 17 completes the set with the single-SiPM WLS-fibre layout (config 4). The absolute efficiencies are the lowest of all four layouts —bare 3.5×10^{-5}, TiO_2 5.6×10^{-4}, Al 1.4×10^{-3}, glossy 1.5×10^{-3}—as expected from the small fibre acceptance, but the reflector ordering and the flattening behaviour are unchanged. The bare plate retains a strong gradient (1.2×10^{-5} at the centre versus 4.0×10^{-5} outside $r = 60\,\mathrm{mm}$, a factor 3.2) and TiO_2 a milder one (3.4×10^{-4} versus 6.0×10^{-4}), whereas Al and glossy are almost flat (the centre and outer values agree to better than 2 %), so even with a single readout face the specular wrappers erase the entry-point dependence.

Across all four layouts the reflector ordering is identical and monotone,

$$\text{bare} \ll TiO_2 < \text{Al} < \text{glossy},$$

and the efficiency scales roughly with the collected photosensitive area, confirming that the wrapper, not the readout layout, governs the collection efficiency.

4.2 Whole-slab uniformity

The full-plate maps also expose an inhomogeneity that the three discrete patches of Sec. 3 can only sample. Because the whole-slab beam irradiates every point of the plate, this inhomogeneity can be quantified *continuously* and with far better statistics.

A purely radial contrast is deliberately *not* used. The read-out layouts have no continuous rotational symmetry—configs 1 and 2 are four-fold symmetric, configs 3 and 4 single-sided—and the plate is square, so the collection efficiency is not a function of radius alone. Beyond the inscribed circle ($r > 125\,\mathrm{mm}$) only the four corner wedges of the plate are populated, so a fixed-width radial bin would mix a shrinking range of directions and its $\varepsilon(r)$ would reflect the geometry of the square rather than the detector. The uniformity is therefore quantified directly in the plate plane (x, y).

Two complementary uniformity measures. The whole-slab scan gives the collection efficiency as a *field* $\varepsilon(x, y)$, the position-binned image of Eq. 3. We characterise this field in two ways. (i) The *normalised map* $\varepsilon/\langle\varepsilon\rangle$, with $\langle\varepsilon\rangle$ the plate-averaged efficiency, is shown for each layout in Figs. 19–22 in 5 mm cells. It is a dimensionless field of unit mean that shows *where* the response is high or low and the spatial pattern of the asymmetry (for instance a top–bottom gradient), but *not* the absolute efficiency. (ii) The *two-dimensional contrast*

$$\kappa_{2\mathrm{D}} = \frac{\max_{\mathrm{cell}}\varepsilon}{\min_{\mathrm{cell}}\varepsilon}$$

collapses the field to a single number, the ratio of the brightest to the darkest region; it answers "how large is the spread" and is the quantity that ranks the layouts and reflectors. κ_{2D} is evaluated on a coarser 5×5 grid of 50 mm cells, because on the 5 mm cells of the maps the per-cell statistics are too thin for a stable max/min—Poisson noise in the extreme cells would inflate it. The two measures therefore use different binning: the contrast quoted below is deliberately the *robust* dynamic range, and is smaller than the raw peak-to-trough range of the fine maps. No symmetry is assumed, so κ_{2D} is meaningful for the single-sided layouts 3 and 4 as well as for the four-sided ones.

The direction-sensitive contrast κ_{2D} ranks the reflectors identically and monotonically in *every* layout,

$$\text{bare} \gg \text{TiO}_2 > \text{Al} \gtrsim \text{glossy},$$

matching the reflector ordering found at the three near-edge patches. Numerically it is much larger than the spread there, because the full plate includes the extreme corners that the three 50 mm patches average away: for the bare plate κ_{2D} runs 9.5, 15.7, 67 and 138 for configs 1–4, against the $\lesssim 4\,\%$ variation seen over the three patches. The specular wrappers, by contrast, stay near unity—$\kappa_{2D} = 1.06$–1.13 for the four-sided layouts 1 and 2 and $\lesssim 1.5$ even for the single-sided layouts 3 and 4—so that for layouts 1 and 2 a muon crossing anywhere on the plate is collected within 13 % of one at the most favourable point, whereas the bare plate loses two orders of magnitude at the corners. TiO_2 sits between the two families ($\kappa_{2D} = 1.9$–16.9). Figure 18 collects all sixteen values as bar charts, one panel per layout: the bare bar rises by more than an order of magnitude from the four-sided to the single-sided layouts, while the reflective bars stay pinned near the unity line. Table 5 lists the same values numerically, resolving the near-unity reflective entries that the linear bar scale cannot.

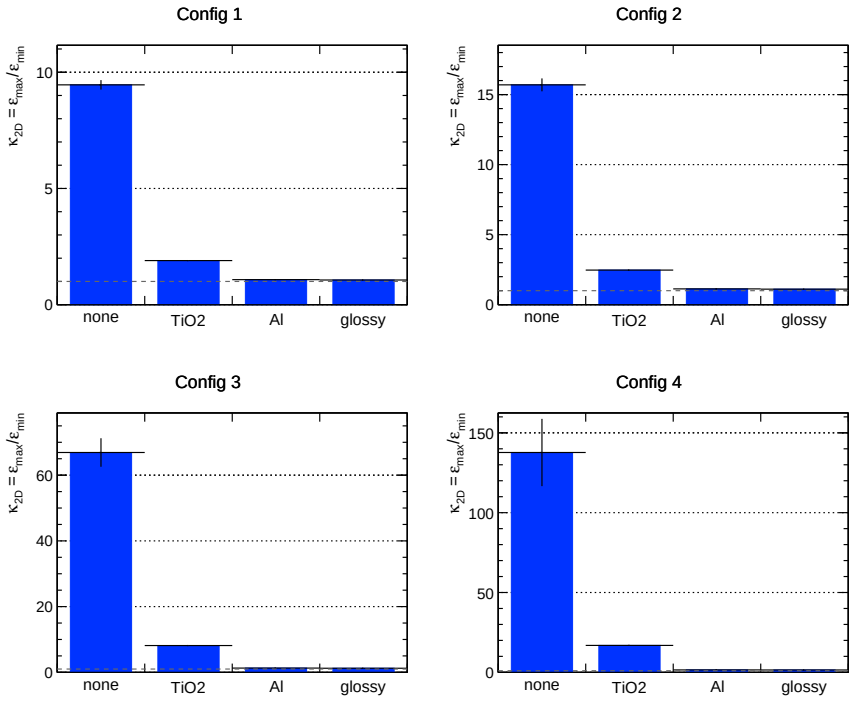


Figure 18: Full-plate position contrast $\kappa_{2D} = \varepsilon_{max}/\varepsilon_{min}$ over the 50 mm cells of the whole-slab map, one panel per layout and one bar per reflector (each panel on its own linear scale). This is the same kind of comparison as the three near-edge patches, but evaluated over the entire plate; the dashed line marks unity (a perfectly uniform response). The bare plate dominates, whereas the specular Al and glossy wrappers stay close to unity and TiO_2 sits in between.

Table 5: Full-plate position contrast κ_{2D} over the 50 mm cells of the whole-slab map, per layout and reflector, with the propagated Poisson uncertainty. These are the values plotted in Fig. 18; the table resolves the near-unity reflective entries that the linear bar scale cannot, and shows the monotonic bare $\gg$ TiO_2 > Al $\gtrsim$ glossy ordering in every layout. The number in parentheses is the uncertainty in the last digits of the preceding value.

Reflector	Config 1	Config 2	Config 3	Config 4
Bare	9.46(20)	15.69(46)	66.9(43)	138(21)
TiO_2	1.90(1)	2.47(1)	8.15(6)	16.9(3)
Al	1.077(2)	1.128(3)	1.293(5)	1.513(13)
Glossy	1.062(2)	1.110(3)	1.245(4)	1.481(12)

The single-sided layouts are the most demanding. For config 4 the two-dimensional contrast is the largest of the entire study ($\kappa_{2D} = 138$): the front–back asymmetry across the plate is far stronger than any radial average reveals. A single-sided layout is simply not described by a radius, so only the full (x, y) map—not a radial profile—carries the information.

Figures 19–22 show the same response in normalised form, $\varepsilon/\langle\varepsilon\rangle$, for each of the four layouts, on a shared *logarithmic* colour scale. A single linear scale cannot hold the range: on the bare plate the response spans several decades about its mean, with the minimum at the corners and the maximum near the edge mid-points, while the reflective wrappers occupy a narrow band around unity. The two four-sided layouts (1 and 2) are nearly azimuthally symmetric, whereas the single-sided layouts (3 and 4) show the one-sided read-out directly, the region opposite the sensors darkening in the bare and TiO_2 panels.

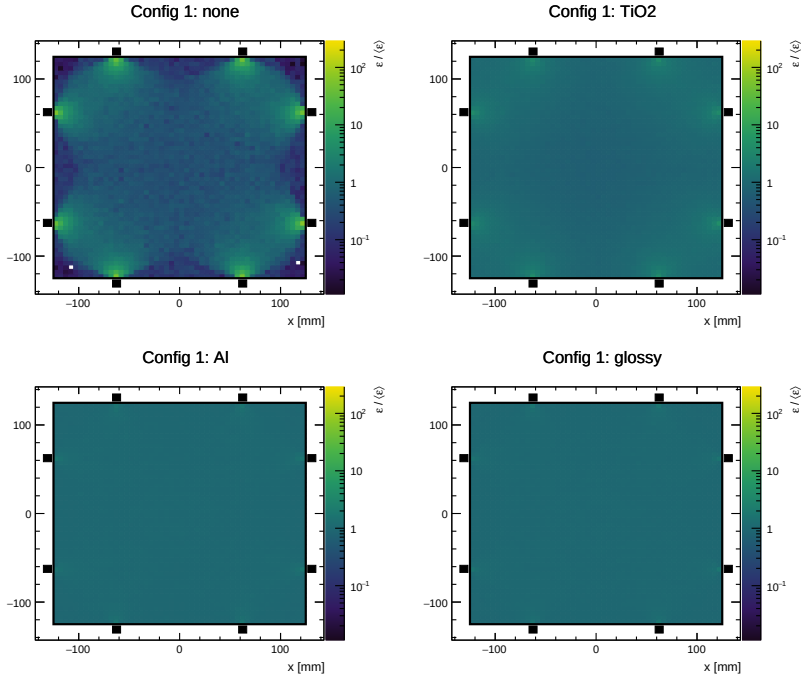


Figure 19: Whole-slab efficiency normalised to its plate average, $\varepsilon/\langle\varepsilon\rangle$, for config 1, one panel per reflector (5 mm cells, the same binning as Figs. 14–17, on a shared *logarithmic* colour scale). The slab outline and the SiPM positions (black squares), or for config 4 the WLS fibre clamp (red bar) with its read-out SiPM, are overlaid as in Fig. 17. The bare plate spans several decades about its mean, with the minimum at the corners and the maximum near the edge mid-points; the reflective wrappers occupy a narrow band around unity, confirming that the position dependence is erased.

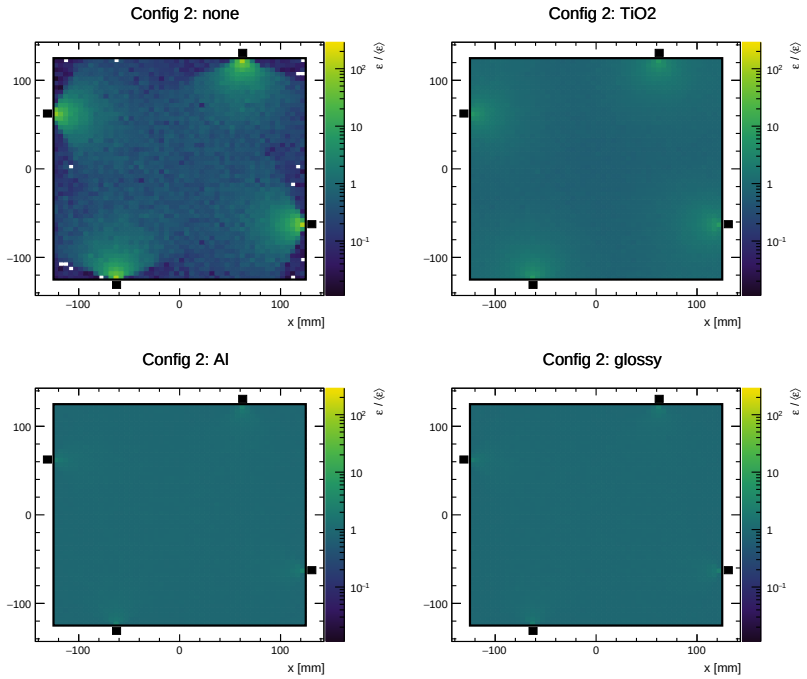


Figure 20: As Fig. 19 but for config 2 (four SiPMs, one per lateral face). The patterns are almost indistinguishable from config 1—the four-sided read-out is effectively azimuthally symmetric—with only the overall normalisation (hence the absolute efficiency) reduced.

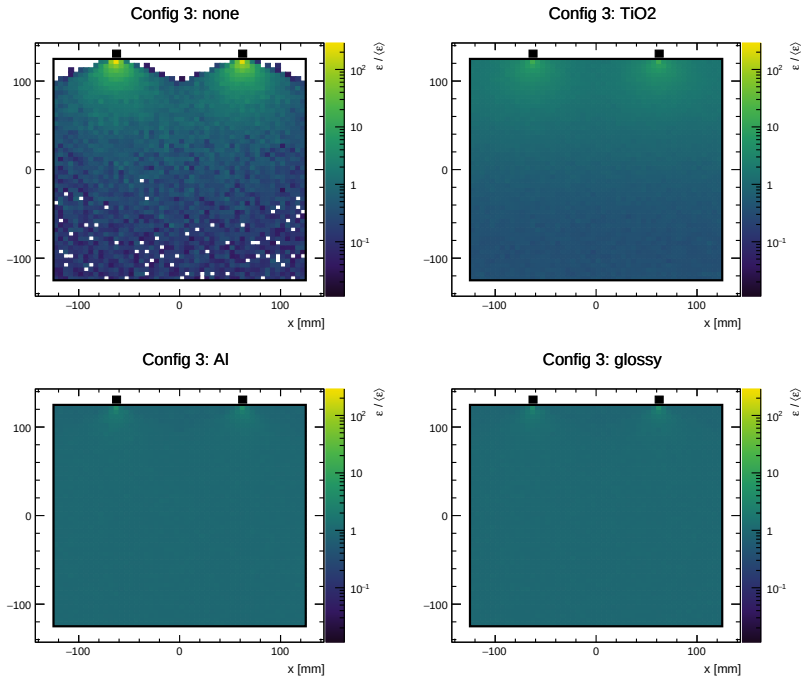


Figure 21: As Fig. 19 but for config 3 (two SiPMs, both on the top $+y$ face). The one-sided read-out breaks the azimuthal symmetry: the bare and TiO_2 panels show a pronounced top–bottom gradient, with the response falling toward the $(-y)$ face opposite the sensors. The specular wrappers close most of this gap.

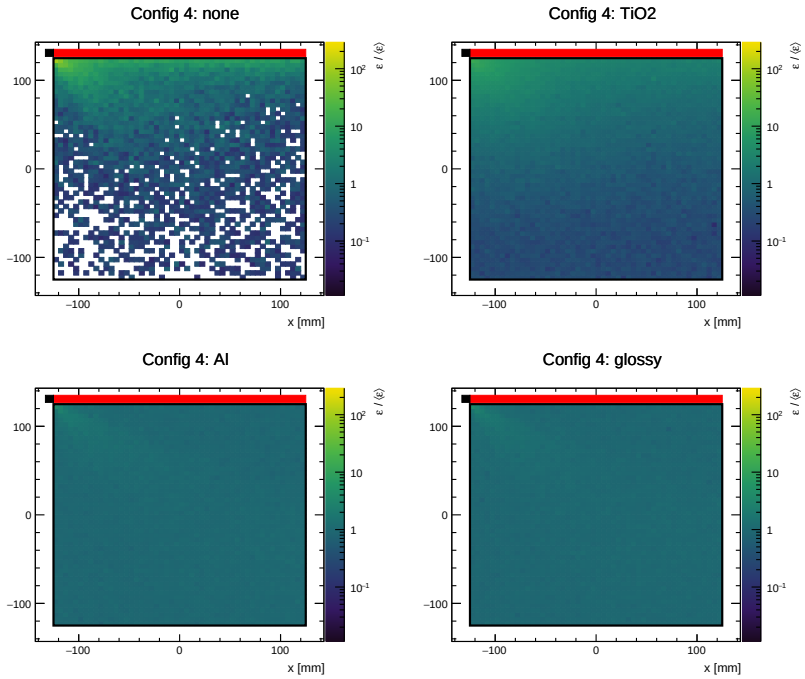


Figure 22: As Fig. 19 but for config 4 (single SiPM + WLS fibre clamp along the top $+y$ edge, read out at the top-left corner). The single fibre read-out produces the strongest asymmetry of all four layouts, the response falling from the instrumented ($+y$) edge toward the opposite ($-y$) side; this is the azimuthal structure that gives config 4 the largest κ_{2D} of the study.

5 Caveats and open items

Intermittent multi-threaded stall. During the matrix production runs a rare stall was observed: on one occasion a single worker thread entered a tight loop (100% CPU) while the remaining threads waited at the event-level barrier, freezing the run. The condition is intermittent—the same macro completed normally on a clean re-run—and is consistent with a pathological optical track (e.g. a photon bouncing in a near-degenerate configuration) rather than a deterministic defect. Because a single stalled task would consume the entire wall-time allocation on a batch system, the production workflow wraps every task in a timeout with automatic retry. The matrix results reported here are unaffected: all 48 files completed (four layouts × four reflectors × three positions), and the frozen run was discarded and regenerated.

Reversed position trend in TiO_2. For the mirror-like wrappers (Al, glossy) the response is essentially flat across the three patches (contrast $<4\,\%$), with the centre marginally the most efficient and the corner the least, consistent with the shortest average path to a sensor. The TiO_2 configuration instead shows a clear corner $>$ edge $>$ centre ordering (a factor 1.7). Since `groundfrontpainted` is purely Lambertian (no TIR, no specular reflection), the sharp corner provides two orthogonal painted walls close to the sensor-bearing edges, and diffusely scattered light is re-emitted into the plate rather than specularly returned; the corner geometry can therefore couple more light into the edge sensors. The effect persists at the higher 1000-event-per-position statistics used here, though no corresponding trend is reported in Ref. [1] and it should be confirmed with independent simulation settings before being interpreted physically.

Scale and normalisation. The absolute efficiencies are model-dependent: they are fixed by the assumed PDE curve, the grease coupling, and the absorbed-fraction of the tabulated reflectivities. The SiPM breakdown-voltage and gain (which drive the PDE) are also temperature-dependent [9],

an effect not modelled here. The robust result is the *ordering* and the qualitative splitting into an effectively absorbing bare case and a reflective family.

6 Summary

A Geant4 optical simulation of a 250 mm×250 mm×10 mm polished plastic scintillator read out by side-mounted SiPMs has been developed and applied to a full matrix of four SiPM layouts and four reflector treatments at three beam positions. The optical physics—unified surface model, finish selection, photocathode PDE via the boundary process, and per-event photon accounting—has been verified against the Geant4 source. The results reproduce the experimentally observed ordering bare $\ll$ TiO_2 $<$ Al $\approx$ glossy, with the bare plate suppressed by two to three orders of magnitude by frustrated total internal reflection into a black holder. A complementary whole-slab irradiation samples the full 250 mm×250 mm plate uniformly and yields a complete collection-efficiency map, which for the bare configuration shows the expected factor-of-three rise from centre to edge. Keeping the three irradiation patches separate—rather than collapsing them to a single beam-axis radius—reveals that the reflective wrappers homogenise the response over the illuminated region: across the three patches the position contrast stays below 4 % in every layout, and the per-sensor light sharing becomes nearly uniform, whereas the bare plate concentrates 91 % of the collected light on the two sensors nearest the beam. Extending the scan to the whole plate sharpens this picture: the full-plate contrast reaches $\kappa_{2D} = 9.5$–138 for the bare plate across the four layouts, against $\kappa_{2D} \lesssim 1.5$ for the specular wrappers.

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